

Qatar-2b

R-1477

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_200302 · created 2026-08-01T12:31:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458910.89933 +/- 0.00094 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.064
Transit depth (Rp/Rs)^2	0.00029 +/- 0.022 [%]
Orbital Inclination (inc)	89.43 +/- 1.69
Airmass coefficient 1 (a1)	0.56 +/- 0.34
Airmass coefficient 2 (a2)	0.38 +/- 0.74
Scatter in the residuals of the lightcurve fit is	56.03 %
Best Comparison Star	None
Optimal Aperture	3.26
Optimal Annulus	4.06
Transit Duration (day)	0.068 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.002 $\pm$ 0.064 vs 0.16327 (deviation 2.52 σ)
Duration fit vs published	0.068 d vs 0.0767 d (deviation 2.8 σ)
Mid-time O-C	-0.3 min
Depth SNR	0.0
Residual scatter	56.03 %

Light curve

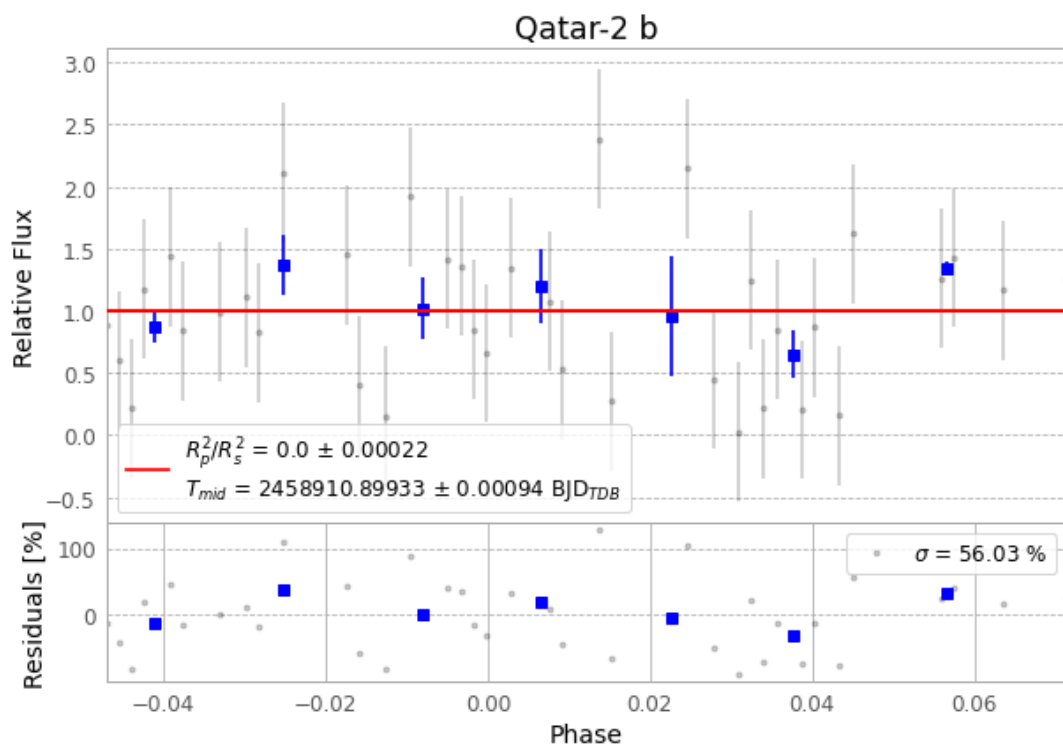


Figure 1 — FinalLightCurve_Qatar-2 b_2020-03-02.png (SHA-256 d8fa0458c678e1d7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [551, 477], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 32bc47af17f930ec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

80 FITS frames (53 MB), Qatar-2200302074812.FITS → Qatar-2200302114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-200302.log (57705d9f87ba...), FinalLightCurve_Qatar-2 b_2020-03-02.png (d8fa0458c678...), FinalLightCurve_Qatar-2 b_2020-03-02.csv (7ce6a62bd02e...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 12:31Z.